

DevOps Engineering Major Project

Major Project Title: DevOpsX 2.0

Objective:

Design and implement a complete CI/CD pipeline integrating code management, build automation, containerization, deployment, and monitoring for a production-grade application.

Modules:

- Source Code Management – Git/GitHub
- Build Automation – Jenkins CI/CD
- Containerization – Docker
- Orchestration – Kubernetes
- Infrastructure as Code – Terraform
- Monitoring – Prometheus & Grafana

Deliverables:

- Fully functional CI/CD pipeline with documentation
- Monitoring dashboard and alerting mechanism
- Infrastructure provisioning scripts (Terraform/Ansible)
- Technical presentation and demonstration video

Industry Relevance:

DevOpsX 2.0 aligns with enterprise practices, emphasizing automation, scalability, and resilience in deployment ecosystems.

Mentor Evaluation Checklist

Criteria	Description	Weightage
Architecture & Integration	Pipeline design and modularity	25%
Automation Quality	Efficiency of build and deploy automation	25%
Scalability & Security	Production readiness and security compliance	25%
Documentation & Presentation	Professional reporting and demo	25%

